PROJECT REPORT

The Design and Simulation of a Digital Data Transmission Error Detection System

Using XOR Gates and Parity Checking

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Abstract

Digital data is transmitted as binary bits (0s and 1s). Noise in a communication channel may change a bit, causing received data to differ from transmitted data. This project demonstrates error detection using XOR gates and parity checking. An XOR network generates an even-parity bit from input data. At the receiver, another XOR network checks the received data together with the parity bit. A checker output of 0 means no parity mismatch is detected; 1 indicates a mismatch. Parity can detect any odd number of bit flips, but it can miss an even number and cannot identify or correct the wrong bit.

1. Introduction

Digital systems exchange information using binary values. During transmission, disturbances can change bits. Error-detection techniques help a receiver identify some corrupted data. Parity checking adds an extra bit; XOR gates are used to generate and check this parity.

2. Objectives

• Design a parity generator using XOR gates.

• Build a parity checker for received data.

• Test correct data and data with a deliberately flipped bit.

• Observe how the checker output changes.

• Understand the advantages and limitations of parity checking.

3. Course Outcomes (COs)

The outcomes below are suggestions based on the uploaded presentation. Replace or relabel them according to the official Course Outcomes supplied by the lecturer.

• CO1 (suggested): Apply Boolean logic and XOR gates.

• CO2 (suggested): Design a digital circuit for a specification.

• CO3 (suggested): Simulate and test circuit behaviour.

• CO4 (suggested): Explain a real-world digital communication use.

4. Components and Tools

• XOR gates (logic ICs or simulator components).

• Binary data inputs and a parity-bit input.

• Logic indicator / output LED.

• Logisim Evolution, Proteus, or equivalent simulator.

• For hardware: power supply, connecting wires, and required logic ICs.

5. Block Diagram and Circuit Design

System flow: Data bits → XOR parity generator → Transmission channel (possible bit flip) → XOR parity checker → Error output.

Parity generator: $P = D2 \oplus D1 \oplus D0$

Parity checker: $S = R2 \oplus R1 \oplus R0 \oplus P$

S = 0 means no parity mismatch is detected; S = 1 means a parity mismatch is detected.

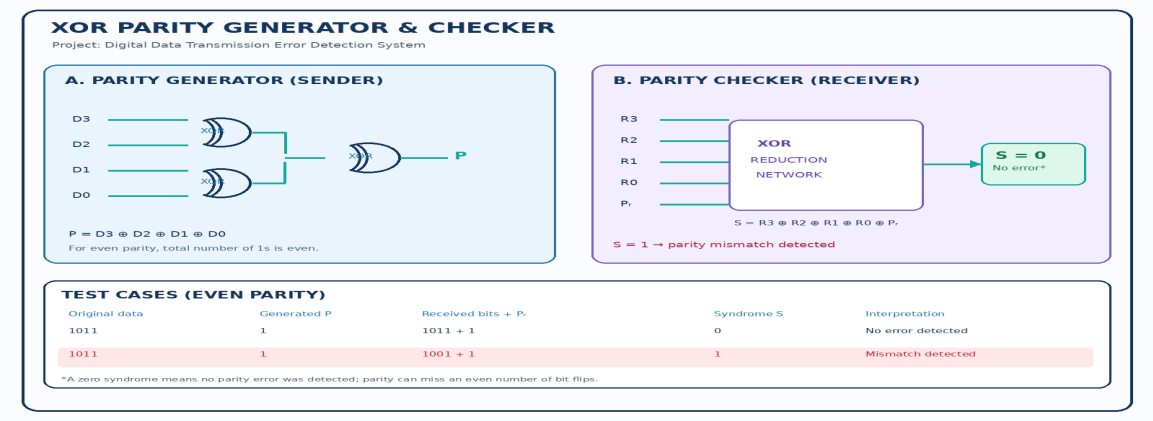


Figure 1. Schematic preview only; verify all wiring in the simulator before treating it as tested.

6. Working Principle

• Enter the binary data bits at the sender.

• XOR the data bits to generate even-parity bit P.

• Transmit data and parity through the channel.

• At the receiver, XOR all received data bits with the received parity bit.

• Read S: 0 indicates no parity mismatch; 1 indicates a parity mismatch.

7. Truth Table

For three data bits, P = D2 ⊕ D1 ⊕ D0. A valid received word has syndrome S = 0.

D2	D1	D0	Parity P	Expected S
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	0
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	0

Illustrative error test cases:

Data sent	Parity P	Received data	Expected result
101	0	101	S = 0; no error detected
101	0	100	S = 1; mismatch detected
110	0	110	S = 0; no error detected
110	0	010	S = 1; mismatch detected

8. Simulation Procedure and Results

These are expected/illustrative results derived from the parity equations, not captured simulator measurements. Run the circuit in Logisim Evolution or another simulator and insert actual screenshots after testing.

• Build the parity generator with XOR gates.

• Connect the generated parity bit to the checking path.

• Build the checker using received data and parity.

• Test a valid word and confirm S = 0.

• Flip one data bit while keeping the parity bit unchanged and confirm S = 1.

• Capture actual screenshots of both tests.

Test	Received data	Parity	Expected result
No bit flip	101	0	S = 0; no error detected
One bit flipped	100	0	S = 1; error detected

9. Advantages

• Simple and low-cost circuit.

• Easy to implement using XOR gates.

• Fast indication of a parity mismatch.

• Useful for learning Boolean logic and digital communication.

10. Limitations

• Cannot identify the incorrect bit.

• Cannot correct corrupted data.

• May miss an even number of bit flips.

• Weaker than stronger error-control methods such as CRC or Hamming codes.

11. Applications and Future Scope

Applications include digital electronics laboratory demonstrations, teaching XOR logic, and introductory demonstrations of parity in data links or memory systems. Future improvements include more data bits, selectable even/odd parity, a visible alarm/LED, and comparison with CRC or Hamming-code methods.

12. Conclusion

This project explains how XOR gates can generate and check an even-parity bit. The checker produces S = 0 when the received word has valid parity and S = 1 when a parity mismatch is detected. The design is simple and useful for demonstrating basic error detection. However, parity checking cannot locate or correct an error and may fail to detect an even number of bit flips. The final report should include verified connections and actual simulation screenshots after testing.

References

• Digital logic and Boolean algebra course notes supplied by the lecturer.

• Logisim Evolution documentation/help, if used for simulation.

• Project presentation: The Design and Simulation of a Digital Data Transmission Error Detection System.

Appendix: GitHub Upload Checklist

• Final project report (PDF and/or DOCX).

• Final PowerPoint presentation.

- Verified circuit/simulation file.
- Circuit diagram and actual simulation screenshots.
- README.md with overview, tools, testing steps, and contributions.
- Poster file (PDF or image).